

Oscillators

Classification, Barkhausen Criterion, RC and LC Oscillators

Ameenudeen PE

Analog Circuits

Unit I: Feedback Amplifiers and Oscillators

Department of Electronics & Communication Engineering

Syllabus covered in this lecture

Prescribed syllabus

Oscillators: Classification, criterion for oscillation, Wien bridge oscillator, RC phase shift oscillator, Hartley and Crystal oscillator.

(Working principle and design equations of the circuits; **analysis of Wien bridge oscillator and RC phase shift oscillator only** is required.)

Treated with full analysis

- RC phase-shift oscillator
- Wien bridge oscillator

Treated qualitatively + design equations

- Hartley oscillator
- Crystal oscillator (Pierce)

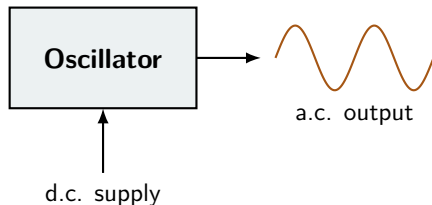
Outline

- 1 What is an Oscillator?
- 2 Classification of Oscillators
- 3 Tank Circuit and the Criterion for Oscillation
- 4 RC Phase Shift Oscillator
- 5 Wien Bridge Oscillator
- 6 Hartley Oscillator
- 7 Crystal Oscillator
- 8 Summary

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The oscillator: a dc-to-ac energy converter



- Any circuit that generates an **alternating output without an externally applied input signal**.
- Energy comes from the dc supply; the circuit converts dc energy into ac energy at a desired frequency.
- The **frequency is fixed by passive components** (L, C, R or a crystal), not by any input.
- Output may be sinusoidal, square, triangular or sawtooth.

Amplifier versus Oscillator

Amplifier

Produces an output whose waveform is similar to the input signal

Energy conversion is *controlled* by the input signal

No input $\Rightarrow$ no energy conversion $\Rightarrow$ no output

Gain and bandwidth set by the circuit

Oscillator

Produces an output signal **without any input signal**

No external signal is needed to maintain the energy conversion

Keeps producing output as long as dc power is present

Frequency set by the passive (frequency-determining) components

An oscillator may be viewed as an amplifier that supplies its own input through a **positive feedback** network.

Why we need them: typical frequency bands

Frequency range	Application
20 Hz – 20 kHz	Audio frequency generator, testing of audio amplifiers
550 kHz – 22 MHz	Radio (AM / SW) broadcasting
88 MHz – 108 MHz	FM radio
1 GHz – 4 GHz	DTH, TV and satellite communication

- Also: clocks for digital systems, carrier generation, local oscillators in receivers, function generators, timing and sweep circuits.

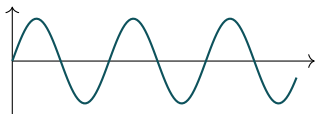
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Classification by output waveform

(i) Sinusoidal (harmonic) oscillators

- Output is a pure sine wave
- Wien bridge, RC phase shift, Hartley, Colpitts, Clapp, crystal



(ii) Non-sinusoidal (relaxation) oscillators

- Square, triangular or sawtooth output
- Multivibrators, UJT relaxation oscillator, 555 timer circuits



Classification by circuit construction

1. **Tuned-circuit (LC) oscillators** — Hartley, Colpitts, Clapp.
Use an LC tank as the frequency-determining network. Practical above ~ 100 kHz.
2. **RC oscillators** — RC phase shift, Wien bridge.
Use resistors and capacitors only. Preferred at audio and low frequencies.
3. **Negative-resistance oscillators** — tunnel diode, UJT relaxation oscillator.
The device's negative-resistance region cancels the circuit losses.
4. **Crystal oscillators** — Pierce, Miller.
A piezoelectric crystal replaces the tank; extremely high frequency stability.
5. **Multivibrators** — astable, monostable, bistable.
Non-sinusoidal output.

Classification by frequency of oscillation

Class	Range	Usual circuit
Audio frequency (AF)	up to 20 kHz	RC phase shift, Wien bridge
Radio frequency (RF)	20 kHz – 30 MHz	Hartley, Colpitts, crystal
Very high frequency (VHF)	30 MHz – 300 MHz	LC, crystal (overtone)
Ultra high frequency (UHF)	300 MHz – 3 GHz	LC with distributed elements
Microwave	above 3 GHz	Cavity, Gunn, IMPATT

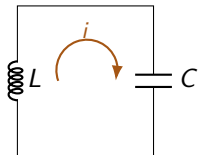
Rule of thumb

At low frequencies the L and C needed for a tank become impractically large and bulky $\Rightarrow$ use **RC** oscillators. At high frequencies stray reactances dominate $\Rightarrow$ use **LC** or **crystal**.

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The LC tank circuit



Energy interchange

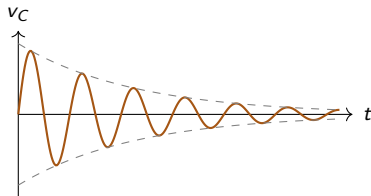
1. C charged from the dc source: energy stored **electrostatically**.
2. C discharges through L ; current builds up slowly (inductor opposes change of current).
3. When C is fully discharged, all energy is **electromagnetic** in the coil.
4. The collapsing field maintains the current and recharges C with *opposite* polarity; the cycle repeats.

This continuous interchange between L and C is the **oscillation**.

$$\text{At resonance } X_L = X_C: \quad 2\pi f_o L = \frac{1}{2\pi f_o C} \implies \boxed{f_o = \frac{1}{2\pi\sqrt{LC}}}$$

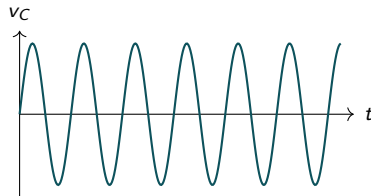
Damped and sustained oscillations

Damped



Practical tank: coil resistance dissipates energy at every interchange, so the amplitude decays.

Sustained (undamped)



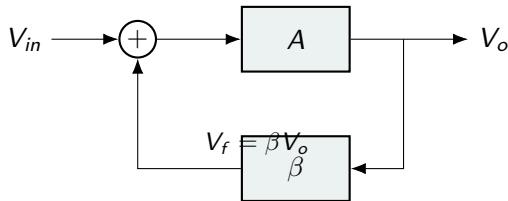
A pulse of energy is returned to the tank *in phase*, once per cycle, by an amplifier through a feedback path — exactly replacing the loss.

The design question

How much energy must be fed back, and with what phase, for the oscillation to be sustained?
⇒ **Barkhausen criterion.**

Criterion for oscillation: Barkhausen criterion

For a **positive** feedback amplifier



$$A_f = \frac{V_o}{V_{in}} = \frac{A}{1 - A\beta}$$

An oscillator has **no external input**, i.e. $V_{in} = 0$, yet $V_o \neq 0$. This is possible only if the denominator vanishes:

$$1 - A\beta = 0 \quad \Rightarrow \quad \boxed{A\beta = 1}$$

$A\beta$ is the **loop gain**.

The two conditions

1. **Magnitude:** $|A\beta| = 1$ (unity loop gain)
2. **Phase:** total phase shift around the loop $\angle A\beta = 0^\circ$ (or 360°)

Barkhausen criterion in practice

How does it start?

- There is no input — oscillation builds up from **circuit noise** and the switch-on transient, which contain all frequencies.
- Only at the frequency where $\angle A\beta = 0^\circ$ does the feedback reinforce; every other frequency is cancelled.
- At start-up the circuit is designed for $|A\beta|$ **slightly greater than 1** so the amplitude grows.

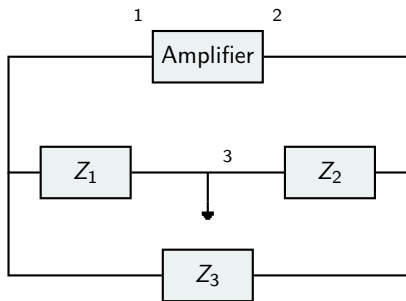
What limits the amplitude?

- If $|A\beta| > 1$ forever, the output would grow until the amplifier clips (distorted output).
- A practical oscillator includes **amplitude limiting**: device non-linearity, a lamp/thermistor, a JFET, or back-to-back diodes, which reduces A as the amplitude rises until $|A\beta| = 1$ exactly.

Summary

$|A\beta| > 1$: amplitude grows • $|A\beta| = 1$: steady sustained oscillation • $|A\beta| < 1$: oscillation dies out

General form of an LC oscillator



Z_1, Z_2, Z_3 are **reactive** elements forming the feedback tank. Z_1 and Z_2 act as a voltage divider, so the feedback fraction is

$$\beta = \frac{Z_1}{Z_2}$$

Oscillator	Z_1	Z_2	Z_3
Hartley	L_1	L_2	C
Colpitts	C_1	C_2	L

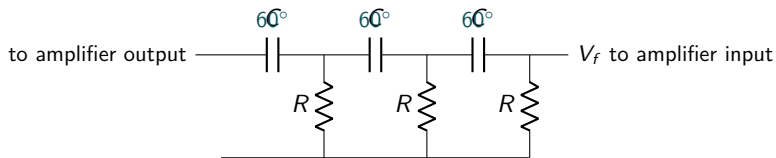
Z_1, Z_2 must be of the *same* kind and Z_3 of the *opposite* kind for the tank to give the required 180° .

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RC phase shift oscillator: the idea

- LC oscillators are impractical at audio frequencies — the required L and C become very bulky.
- In an RC oscillator the 180° needed from the feedback network is produced by **cascaded RC sections** instead of a tank.
- A common-emitter amplifier (or an inverting op-amp) supplies the other 180° , making the loop phase 360° .



Why three sections?

For one high-pass RC section the phase lead is

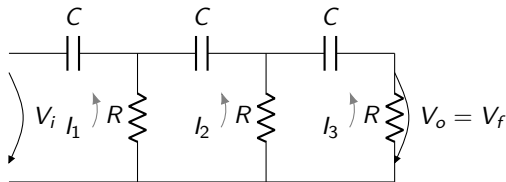
$$\phi = \tan^{-1}\left(\frac{1}{\omega CR}\right), \quad 0^\circ < \phi < 90^\circ$$

- 90° is approached only as $\omega \rightarrow 0$, where the section also gives zero output. So one section can never give a usable 90° .
- **Two** sections could give 180° only in the limit $\phi = 90^\circ$ each — not realisable.
- **Three** sections, each contributing 60° at one particular frequency, give the required 180° with adequate output.
- Four or more sections work too, but need more components for no benefit.

Therefore

Choose R and C so that each section gives exactly 60° at the desired frequency f_o . The ladder then produces 180° ; the CE stage adds 180° ; loop phase = 360° and the Barkhausen phase condition is satisfied **at that single frequency only**.

Analysis: loop equations of the RC ladder



Applying KVL to the three loops:

$$\begin{aligned} I_1 \left(R + \frac{1}{j\omega C} \right) - I_2 R &= V_i \\ -I_1 R + I_2 \left(2R + \frac{1}{j\omega C} \right) - I_3 R &= 0 \\ -I_2 R + I_3 \left(2R + \frac{1}{j\omega C} \right) &= 0 \end{aligned}$$

Analysis: solving for the feedback factor

Replacing $j\omega$ by s and writing in matrix form,

$$\begin{bmatrix} R + \frac{1}{sC} & -R & 0 \\ -R & 2R + \frac{1}{sC} & -R \\ 0 & -R & 2R + \frac{1}{sC} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_i \\ 0 \\ 0 \end{bmatrix}$$

By Cramer's rule,

$$I_3 = \frac{V_i s^3 R^2 C^3}{1 + 5sRC + 6s^2 R^2 C^2 + s^3 R^3 C^3}$$

Since $V_o = V_f = I_3 R$,

$$\beta = \frac{V_f}{V_i} = \frac{s^3 R^3 C^3}{1 + 5sRC + 6s^2 R^2 C^2 + s^3 R^3 C^3}$$

Analysis: the frequency of oscillation

Put $s = j\omega$:

$$\beta = \frac{-j\omega^3 R^3 C^3}{1 + 5j\omega RC - 6\omega^2 R^2 C^2 - j\omega^3 R^3 C^3}$$

Divide numerator and denominator by $-j\omega^3 R^3 C^3$ and let $\alpha = \frac{1}{\omega RC}$:

$$\beta = \frac{1}{1 + 6j\alpha - 5\alpha^2 - j\alpha^3} = \frac{1}{(1 - 5\alpha^2) + j\alpha(6 - \alpha^2)}$$

For a phase shift of 180° the **imaginary part of the denominator must be zero**:

$$\alpha(6 - \alpha^2) = 0 \implies \alpha^2 = 6 \implies \alpha = \sqrt{6}$$

$$\frac{1}{\omega RC} = \sqrt{6} \implies \omega = \frac{1}{RC\sqrt{6}} \implies \boxed{f_o = \frac{1}{2\pi RC\sqrt{6}}}$$

Analysis: the gain condition

At $\alpha^2 = 6$ the feedback factor becomes

$$\beta = \frac{1}{1 - 5\alpha^2} = \frac{1}{1 - 5(6)} = -\frac{1}{29} \quad \Rightarrow \quad \boxed{|\beta| = \frac{1}{29}}$$

The minus sign confirms the 180° phase shift of the network.

For oscillation, $|A\beta| \geq 1$:

$$|A| |\beta| \geq 1 \quad \Rightarrow \quad |A| \geq \frac{1}{|\beta|} \quad \Rightarrow \quad \boxed{|A| \geq 29}$$

Design rule

The amplifier must have a voltage gain of **at least 29** (and must invert). With an op-amp in inverting mode: $\frac{R_f}{R_1} = 29$, i.e. $R_f = 29R_1$.

RC phase shift oscillator: design procedure

Design steps

1. Fix the required f_o .
2. Choose a convenient C (typically 100 pF – 0.1 μ F).
3. Compute R from $R = \frac{1}{2\pi f_o C \sqrt{6}}$.
4. Design the amplifier for $|A| \geq 29$; with an op-amp take $R_f = 29R_1$ (use a slightly larger value, e.g. $R_f \approx 30R_1$, to guarantee start-up).

Watch the loading

In the BJT version the amplifier input resistance h_{ie} appears in parallel with the last R , so the network is no longer three identical sections. The required gain then works out higher than 29 — design for a comfortable margin.

RC phase shift oscillator: worked examples

Example 1

In an RC phase shift oscillator $R_1 = R_2 = R_3 = 200 \text{ k}\Omega$ and $C_1 = C_2 = C_3 = 100 \text{ pF}$. Find f_o .

$$f_o = \frac{1}{2\pi RC\sqrt{6}} = \frac{1}{2\pi(200 \times 10^3)(100 \times 10^{-12})(2.449)} = \mathbf{3.248 \text{ kHz}}$$

Example 2

An RC phase shift oscillator is to oscillate at 955 Hz with $R = 680 \text{ k}\Omega$.

$$C = \frac{1}{2\pi f_o R\sqrt{6}} = \frac{1}{2\pi(955)(680 \times 10^3)(2.449)} = \mathbf{100 \text{ pF}}$$

RC phase shift oscillator: features

Advantages

- No bulky inductors; simple and cheap.
- Good sine wave output.
- Suited to the audio range (a few Hz to ~ 200 kHz).

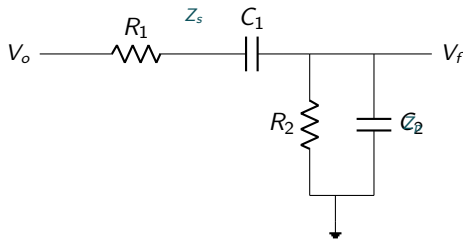
Limitations

- Frequency cannot be varied easily — all three R 's (or C 's) must be changed *simultaneously and identically*.
- Needs a large gain (≥ 29); the network attenuates heavily.
- Poor frequency stability compared with LC or crystal circuits.
- In the BJT version the input resistance h_{ie} loads the last section, so the practical gain needed is somewhat higher than 29.

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Wien bridge oscillator: the lead-lag network



- Series arm $Z_s = R_1 + \frac{1}{j\omega C_1}$ — a **high-pass** (lead) section.
- Shunt arm $Z_p = R_2 \parallel \frac{1}{j\omega C_2}$ — a **low-pass** (lag) section.
- At low f the series capacitor blocks; at high f the shunt capacitor shorts. In between there is a **single frequency of maximum output with exactly zero phase shift**.
- Because the network gives 0° , the amplifier must also give 0° — a **non-inverting** amplifier.

Analysis of the Wien bridge network

Take $R_1 = R_2 = R$ and $C_1 = C_2 = C$. Then

$$Z_s = R + \frac{1}{j\omega C} = \frac{1 + j\omega RC}{j\omega C}, \quad Z_p = \frac{R}{1 + j\omega RC}$$

The network is a voltage divider:

$$\beta = \frac{V_f}{V_o} = \frac{Z_p}{Z_s + Z_p} = \frac{\frac{R}{1 + j\omega RC}}{\frac{1 + j\omega RC}{j\omega C} + \frac{R}{1 + j\omega RC}}$$

Multiplying numerator and denominator by $j\omega C(1 + j\omega RC)$:

$$\beta = \frac{j\omega RC}{(1 + j\omega RC)^2 + j\omega RC} = \frac{j\omega RC}{(1 - \omega^2 R^2 C^2) + 3j\omega RC}$$

Analysis: frequency and gain conditions

Dividing numerator and denominator by $j\omega RC$:

$$\beta = \frac{1}{3 + j\left(\omega RC - \frac{1}{\omega RC}\right)}$$

The phase shift is zero only when the imaginary term vanishes:

$$\omega RC = \frac{1}{\omega RC} \implies \omega_o^2 = \frac{1}{R^2 C^2} \implies \omega_o = \frac{1}{RC} \implies f_o = \frac{1}{2\pi RC}$$

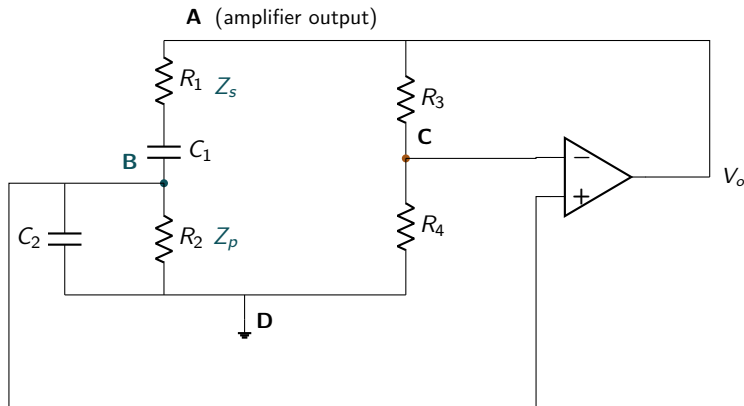
At this frequency

$$\beta = \frac{1}{3} \implies |A| \geq \frac{1}{|\beta|} = \boxed{3}$$

Design rule

A **non-inverting** amplifier of gain exactly 3 is required: $1 + \frac{R_3}{R_4} = 3 \Rightarrow \boxed{R_3 = 2R_4}$ (the two resistive arms of the bridge)

The complete Wien bridge oscillator



Four arms: $Z_s = R_1 + \frac{1}{j\omega C_1}$ (A-B), $Z_p = R_2 \parallel \frac{1}{j\omega C_2}$ (B-D), R_3 (A-C) and R_4 (C-D). The amplifier output drives the diagonal **A-D**; the amplifier *inputs* are the other diagonal **B-C**. R_3 and R_4 are simultaneously the bridge's resistive arms **and** the amplifier's gain-setting network.

Why a bridge? Balance *is* the Barkhausen condition

The amplifier senses the **difference** between the two bridge taps:

$$V_+ = V_o \frac{Z_p}{Z_s + Z_p} = \beta V_o \quad (\text{frequency selective})$$

$$V_- = V_o \frac{R_4}{R_3 + R_4} \quad (\text{frequency independent})$$

Feedback through R_3 – R_4 makes the closed-loop gain $A = 1 + \frac{R_3}{R_4}$, so the Barkhausen condition $A\beta = 1$ requires

$$\beta = \frac{1}{A} = \frac{R_4}{R_3 + R_4} \implies \frac{Z_p}{Z_s + Z_p} = \frac{R_4}{R_3 + R_4} \implies \boxed{R_3 Z_p = R_4 Z_s}$$

The last step is just cross-multiplication: $Z_p(R_3 + R_4) = R_4(Z_s + Z_p)$, in which the $R_4 Z_p$ terms cancel.

That is exactly the balance condition of the bridge — the two dividers deliver equal voltages, so the detector diagonal B–C carries nothing.

Why a bridge? (continued)

The reason we may equate the two arms

$V_+ = V_-$ means the bridge is balanced. So the statement “the oscillator works at the frequency at which the Wien bridge is balanced” and the statement “ $A\beta = 1$ ” are **the same equation**, written two ways. The frequency condition and the gain condition therefore fall out of one complex equation — one from its real part, one from its imaginary part.

In practice

At exact balance the net input is zero, so a real oscillator is run *marginally* off balance: R_3/R_4 is set slightly above 2 so that oscillation can start. The AGC element then pulls the bridge back towards balance as the amplitude grows.

Why this gives a clean sine wave

Close to balance the net signal is a small difference of two large quantities, so its *phase* swings very rapidly with frequency. The circuit is therefore far more frequency-selective than the β curve alone suggests — hence the low distortion and good stability.

Bridge balance: complete derivation

Substituting $Z_p = \frac{R_2}{1 + j\omega C_2 R_2}$ and $Z_s = R_1 + \frac{1}{j\omega C_1} = R_1 - \frac{j}{\omega C_1}$ into $R_3 Z_p = R_4 Z_s$:

$$R_3 \left[\frac{R_2}{1 + j\omega C_2 R_2} \right] = R_4 \left[R_1 - \frac{j}{\omega C_1} \right]$$

Multiplying both sides by $(1 + j\omega C_2 R_2)$:

$$R_2 R_3 = R_4 (1 + j\omega C_2 R_2) \left(R_1 - \frac{j}{\omega C_1} \right)$$

Expanding the product (note $j \cdot (-j) = +1$):

$$R_2 R_3 = R_4 \left[R_1 + \frac{C_2 R_2}{C_1} \right] + j R_4 \left[\omega C_2 R_2 R_1 - \frac{1}{\omega C_1} \right]$$

The left-hand side is purely real, so the real and imaginary parts must match separately.

Bridge balance: the two conditions

Imaginary part = 0 $\rightarrow$ frequency of oscillation

$$\omega C_2 R_2 R_1 - \frac{1}{\omega C_1} = 0 \implies \omega^2 = \frac{1}{C_1 C_2 R_1 R_2} \implies f_o = \frac{1}{2\pi \sqrt{C_1 C_2 R_1 R_2}}$$

$$R_1 = R_2 = R, C_1 = C_2 = C \implies \boxed{f_o = \frac{1}{2\pi RC}}$$

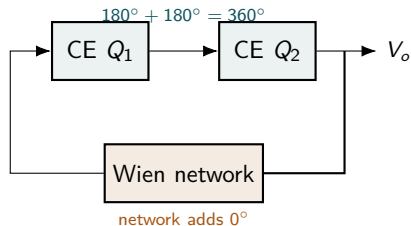
Real part $\rightarrow$ gain condition

$$R_2 R_3 = R_4 R_1 + \frac{C_2}{C_1} R_2 R_4 \xrightarrow{\div R_2 R_4} \frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

$$R_1 = R_2 = R, C_1 = C_2 = C \implies \boxed{R_3 = 2R_4} \implies A = 1 + \frac{R_3}{R_4} = 3$$

Identical to the loop-gain result ($f_o = 1/2\pi RC$, $|\beta| = 1/3$, $|A| \geq 3$) — as it must be, since the two routes solve the same equation.

Transistor realisation



- The network needs a **non-inverting** amplifier, so a single CE stage will not do — it inverts.
- Two CE stages in cascade give $180^\circ + 180^\circ = 360^\circ$, which is equivalent to 0° .
- The first stage's emitter resistor takes the place of R_4 , so the same two resistors again set both the bridge ratio and the gain.
- An op-amp in non-inverting mode does the same job in one device and is the usual modern choice.

Amplitude stabilisation — why it matters

- The gain condition is a **knife edge**: $A = 3$ exactly, i.e. the bridge must sit at balance.
- $A < 3$: oscillation dies. $A > 3$: the output grows until the amplifier saturates and the sine wave is clipped.
- Component tolerance and temperature drift make an exact 3 impossible to hold with fixed resistors.

Practical solutions (automatic gain control)

- **Tungsten lamp** or **thermistor** in place of R_4 — as the amplitude rises, the filament heats, its resistance rises, and the gain falls back to 3. (Used in the original HP200A.)
- **JFET** used as a voltage-controlled resistance in the feedback path.
- **Back-to-back diodes** across R_3 to compress the gain at large amplitudes — simplest, slight distortion.

Wien bridge oscillator: design procedure

Design steps

1. Fix f_o ; choose a convenient C ; compute $R = \frac{1}{2\pi f_o C}$.
2. Set the amplifier gain to 3: $R_3 = 2R_4$ (add AGC as above).
3. For a variable-frequency generator, gang the two R 's (or switch the two C 's for range changing).

Why it is the workhorse of audio signal generators

Only two elements set the frequency, so a single ganged potentiometer sweeps a whole decade; switching C changes range. Compare with the RC phase-shift oscillator, where three elements must track each other exactly.

Wien bridge oscillator: worked examples

Example 1

$R = 100 \text{ k}\Omega$, $f_o = 10 \text{ kHz}$. Find C .

$$C = \frac{1}{2\pi f_o R} = \frac{1}{2\pi(10^4)(10^5)} = \mathbf{159 \text{ pF}}$$

Example 2

$C_1 = C_2 = 0.001 \text{ }\mu\text{F}$, $R_1 = 10 \text{ k}\Omega$ fixed, R_2 variable. Find the range of R_2 for $f = 10\text{--}50 \text{ kHz}$, using $f = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$, i.e. $R_2 = \frac{1}{(2\pi f)^2 C^2 R_1}$.

$$f = 10 \text{ kHz} \Rightarrow R_2 = \mathbf{25.33 \text{ k}\Omega}, \quad f = 50 \text{ kHz} \Rightarrow R_2 = \mathbf{1.013 \text{ k}\Omega}$$

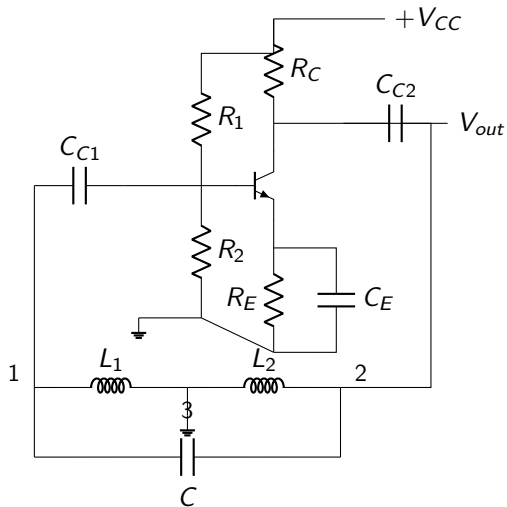
Wien bridge versus RC phase shift

	RC phase shift	Wien bridge
Feedback network	3 cascaded RC sections	Lead-lag (series RC + parallel RC)
Phase shift of network	180°	0°
Amplifier	Inverting (one CE stage)	Non-inverting (two CE stages / op-amp)
β at f_o	$1/29$	$1/3$
Minimum gain	29	3
Frequency	$f_o = \frac{1}{2\pi RC\sqrt{6}}$	$f_o = \frac{1}{2\pi RC}$
Tuning	Difficult (3 elements)	Easy (gang 2 elements)
Typical use	Fixed audio frequency	Audio signal generators (up to ~ 1 MHz)

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Hartley oscillator: circuit



Hartley oscillator: construction and working

Construction

- Comparing with the general form: $Z_1 = L_1$, $Z_2 = L_2$ (both inductive), $Z_3 = C$ (capacitive).
- R_1, R_2, R_E give the dc bias; C_E is the emitter bypass; C_{C1}, C_{C2} are coupling capacitors.
- The tank (L_1, L_2, C) is the frequency-determining network; the coil is **centre-tapped** at terminal 3, which is grounded.

Working

- On switch-on the transient sets up damped oscillation in the tank.
- Terminal 3 is at ground potential, so when terminal 1 is positive, terminal 2 is negative: the tapped coil provides **180°** between terminals 1 and 2.
- The CE amplifier provides a further **180°** .
- Total loop phase = 360° ; if the feedback is adjusted so that $A\beta = 1$, sustained oscillation results.

Hartley oscillator: design equations

Frequency of oscillation

$$f_o = \frac{1}{2\pi\sqrt{L_{eq}C}} \quad \text{where} \quad L_{eq} = L_1 + L_2 + 2M$$

M is the mutual inductance between the two coil sections (if the coils are not coupled, $L_{eq} = L_1 + L_2$).

Feedback fraction and gain condition

Output appears across L_2 , feedback is taken across L_1 :

$$\beta = \frac{Z_1}{Z_2} = \frac{X_{L_1}}{X_{L_2}} = \frac{L_1}{L_2} \quad \Rightarrow \quad |A| \geq \frac{1}{\beta} = \frac{L_2}{L_1}$$

Hartley oscillator: worked example

Example

$C = 500 \text{ pF}$, $L_1 = 12 \text{ } \mu\text{H}$, $L_2 = 38 \text{ } \mu\text{H}$ (no mutual coupling).

$$L_{eq} = 50 \text{ } \mu\text{H}, \quad f_o = \frac{1}{2\pi\sqrt{(50 \times 10^{-6})(500 \times 10^{-12})}} = \mathbf{1.0 \text{ MHz}}, \quad \beta = \frac{12}{38} = \mathbf{0.316}$$

Hartley and its dual: the Colpitts oscillator

	Hartley	Colpitts
Z_1, Z_2 (divider)	Two inductors L_1, L_2	Two capacitors C_1, C_2
Z_3	Capacitor C	Inductor L
Frequency	$f_o = \frac{1}{2\pi\sqrt{L_{eq}C}}, L_{eq} = L_1 + L_2 + 2M$	$f_o = \frac{1}{2\pi\sqrt{LC_{eq}}}, C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$
Feedback fraction	$\beta = \frac{L_1}{L_2}$	$\beta = \frac{C_2}{C_1}$
Tuning	Single variable C — easy	Needs ganged capacitors

- Both are used from about 100 kHz to tens of MHz.
- Hartley is convenient for wide tuning ranges (single tuning capacitor); Colpitts gives a purer waveform at high frequency because the capacitive divider shunts harmonics.

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- 7 Crystal Oscillator**

Why a crystal? The piezoelectric effect

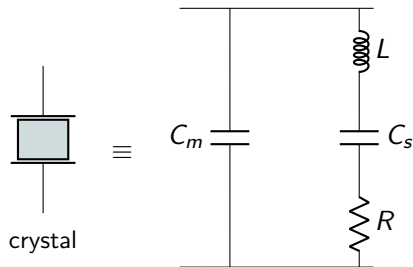
- Some natural crystals show the **piezoelectric effect**: an applied ac voltage makes them vibrate mechanically, and mechanical stress generates a voltage.
- The mechanical resonance is extremely sharp, so the crystal behaves as a tuned circuit of **very high Q** (often 10^4 – 10^6 , against ~ 100 for an LC tank).
- Consequence: frequency stability of a few parts per million, essentially independent of supply voltage, temperature and transistor parameters.

Materials

- **Rochelle salt** — greatest piezoelectric activity, but mechanically weak; used in microphones, headsets, loudspeakers.
- **Tourmaline** — least active but strongest; expensive, very high frequencies.
- **Quartz** — the compromise: cheap, available, rugged. **Used for RF oscillators and filters.**

Fundamental frequency varies inversely with the thickness t of the slab: $f \propto 1/t$ — thinner crystal, higher frequency (and more fragile).

AC equivalent circuit of a crystal



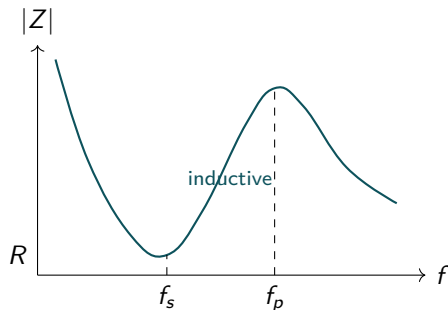
- C_m — **mounting capacitance** of the two metal plates holding the slab (present even when not vibrating).
- L, C_s, R — the **motional** (vibrational) branch: L is the mass of the crystal, C_s its compliance, R its frictional loss.

Typical values

L : henries C_s : fractions of a pF R : hundreds of Ω C_m : a few pF

$$Q = \frac{\omega_s L}{R} = \frac{1}{\omega_s C_s R} \text{ — very large}$$

Series and parallel resonance



Series resonance f_s

The motional branch resonates, $X_L = X_{C_s}$; the crystal presents a very **low** impedance $Z = R$.

$$f_s = \frac{1}{2\pi\sqrt{LC_s}}$$

Parallel (anti-)resonance f_p

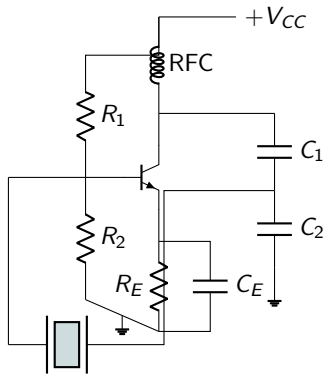
The series leg (now inductive) resonates with C_m ; the crystal presents a very **high** impedance.

$$f_p = \frac{1}{2\pi\sqrt{LC_p}}, \quad C_p = \frac{C_m C_s}{C_m + C_s}$$

$C_p < C_s$, so $f_p > f_s$ always — and they lie very close together.

Between f_s and f_p the crystal is inductive.

Pierce crystal oscillator



- A Colpitts oscillator in which the inductor is replaced by the crystal, operating just above f_s where it is inductive.
- R_1, R_2, R_E set the dc bias; C_E bypasses R_E ; the RFC provides the dc collector path while blocking ac from the supply.
- C_1, C_2 form the capacitive divider, so $\beta = C_2/C_1$ as in a Colpitts circuit.
- The frequency is set almost entirely by the crystal, not by C_1, C_2 — hence the excellent stability.

Merits

Very high stability; unaffected by supply, temperature and transistor parameter changes. *Limitation:* frequency is essentially fixed.

Crystal oscillator: worked example

Problem

A crystal has $L = 0.4 \text{ H}$, $C_s = 0.085 \text{ pF}$, $C_m = 1 \text{ pF}$ and $R = 5 \text{ k}\Omega$. Find (i) the series resonant frequency, (ii) the parallel resonant frequency and (iii) the Q factor.

(i) Series resonance

$$f_s = \frac{1}{2\pi\sqrt{LC_s}} = \frac{1}{2\pi\sqrt{(0.4)(0.085 \times 10^{-12})}} = \mathbf{863 \text{ kHz}}$$

(ii) Parallel resonance

$$C_p = \frac{C_m C_s}{C_m + C_s} = \frac{(1)(0.085)}{1.085} = 0.0783 \text{ pF}, \quad f_p = \frac{1}{2\pi\sqrt{LC_p}} = \mathbf{899 \text{ kHz}}$$

(iii) Quality factor

$$Q = \frac{2\pi f_s L}{R} = \frac{2\pi(863 \times 10^3)(0.4)}{5 \times 10^3} = \mathbf{434}$$

Note how close f_s and f_p are — about 4% apart here; in practice the separation is usually much smaller.

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Comparison of the four oscillators

	RC phase shift	Wien bridge	Hartley	Crystal
Frequency-setting	3 RC sections	Lead-lag RC	Tapped $L + C$	Quartz crystal
Network phase	180°	0°	180°	$\approx 0^\circ$ (Pierce)
Amplifier	Inverting	Non-inverting	Inverting (CE)	Inverting (CE)
β	$1/29$	$1/3$	L_1/L_2	C_2/C_1
Minimum gain	29	3	L_2/L_1	C_1/C_2
f_o	$\frac{1}{2\pi RC\sqrt{6}}$	$\frac{1}{2\pi RC}$	$\frac{1}{2\pi\sqrt{L_{eq}C}}$	$\frac{1}{2\pi\sqrt{LC_s}}$
Range	AF	AF to ~ 1 MHz	100 kHz–tens of MHz	kHz–MHz (fixed)
Stability	Poor	Good	Moderate	Excellent

Formula sheet (1 of 2): criterion and RC oscillators

Criterion for oscillation

$$A\beta = 1; \quad |A\beta| = 1$$

$$\angle A\beta = 0^\circ \text{ or } 360^\circ$$

RC phase shift

$$f_o = \frac{1}{2\pi RC\sqrt{6}}, \quad |\beta| = \frac{1}{29}, \quad |A| \geq 29$$

Tank circuit

$$f_o = \frac{1}{2\pi\sqrt{LC}}$$

Wien bridge

$$f_o = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \xrightarrow{\text{equal}} \frac{1}{2\pi RC}$$

$$\frac{R_3}{R_4} = \frac{C_2}{C_1} + \frac{R_1}{R_2}; \quad |\beta| = \frac{1}{3}, \quad |A| \geq 3$$

Formula sheet (2 of 2): LC and crystal oscillators

Hartley

$$f_o = \frac{1}{2\pi\sqrt{L_{eq}C}}, \quad L_{eq} = L_1 + L_2 + 2M$$
$$\beta = \frac{L_1}{L_2}, \quad |A| \geq \frac{L_2}{L_1}$$

Crystal

$$f_s = \frac{1}{2\pi\sqrt{LC_s}}, \quad f_p = \frac{1}{2\pi\sqrt{LC_p}}$$
$$C_p = \frac{C_m C_s}{C_m + C_s}, \quad Q = \frac{\omega_s L}{R}$$
$$f_p > f_s \text{ always}$$

Colpitts

$$f_o = \frac{1}{2\pi\sqrt{LC_{eq}}}, \quad C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$
$$\beta = \frac{C_2}{C_1}$$

Phase per RC section

$$\phi = \tan^{-1}\left(\frac{1}{\omega CR}\right)$$

Tutorial problems

1. In a Hartley oscillator $L_2 = 0.4 \text{ mH}$ and $C = 0.004 \mu\text{F}$. If $f_o = 120 \text{ kHz}$, find L_1 (neglect mutual inductance). *(0.04 mH)*
2. In a transistorised Hartley oscillator the two inductors are 2 mH and $20 \mu\text{H}$, and the frequency is to be varied from 950 kHz to 2050 kHz . Find the range of C . *(2.98 pF – 13.89 pF)*
3. In a Hartley oscillator $C = 500 \text{ pF}$ and the coil sections are $12 \mu\text{H}$ and $38 \mu\text{H}$. Find f_o and β . *(1 MHz, 0.316)*
4. In an RC phase shift oscillator $R_1 = R_2 = R_3 = 200 \text{ k}\Omega$ and $C_1 = C_2 = C_3 = 100 \text{ pF}$. Find f_o . *(3.248 kHz)*
5. In an RC phase shift oscillator $f_o = 955 \text{ Hz}$ and $R_1 = R_2 = R_3 = 680 \text{ k}\Omega$. Find C . *(100 pF)*
6. In a Wien bridge oscillator $R = 100 \text{ k}\Omega$ and $f_o = 10 \text{ kHz}$. Find C . *(159 pF)*
7. A Wien bridge uses $C_1 = C_2 = 0.001 \mu\text{F}$ and $R_1 = 10 \text{ k}\Omega$, with R_2 variable. Find R_2 for $f = 10 \text{ kHz}$ to 50 kHz . *(1.013 kΩ – 25.33 kΩ)*
8. A crystal has $L = 2 \text{ H}$, $C_s = 0.01 \text{ pF}$, $R = 2 \text{ k}\Omega$ and $C_m = 2 \text{ pF}$. Find f_s and f_p . *(1.125 MHz, 1.128 MHz)*

Key take-aways

1. An oscillator is an amplifier with **positive feedback** that supplies its own input; it converts dc to ac.
2. **Barkhausen criterion**: $|A\beta| = 1$ and $\angle A\beta = 0^\circ$. Start-up needs $|A\beta| > 1$, amplitude limiting then brings it to unity.
3. The **phase condition fixes the frequency**; the **magnitude condition fixes the required gain**. Every derivation in this unit is an application of these two statements.
4. RC networks $\rightarrow$ audio frequencies; LC tanks $\rightarrow$ radio frequencies; crystals $\rightarrow$ maximum stability.
5. Remember the pairs: $(1/29, 29)$ for RC phase shift and $(1/3, 3)$ for Wien bridge.

Questions?